
Finite- K Posterior Learning with One-Draw Primal Updates

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Abstract

1 Multi-sample reasoning generates several candidate solutions and selects among
2 them at inference time. Reasoning models use training objectives defined for a
3 single solver execution, although deployment evaluates a candidate pool under a
4 fixed budget. We place a regularized posterior over solver parameters, using each
5 posterior sample as one candidate solver. For a prescribed pool size K and rea-
6 soning depth, we minimize the probability that all K candidates fail. An unbiased
7 stochastic estimate of this finite- K objective requires K solver executions per up-
8 date. To address this, we derive an exact Fenchel representation whose primal
9 update uses one posterior draw given a per-problem dual state. We also track this
10 state with a closed-form Bregman update and one running failure estimate per
11 problem. Together, this targets at least one successful candidate among K poste-
12 rior samples using one posterior draw per update. Across challenging reasoning
13 benchmarks, we validate finite- K posterior learning improves multi-sample per-
14 formance.

15 1 Introduction

16 2 Related work

17 2.1 Multi-sample reasoning and test-time scaling

18 2.2 Posterior learning for reasoning

19 2.3 Efficient optimization of multi-sample objectives

20 3 Finite- K posterior learning

21 We formulate posterior learning under the prescribed inference protocol as a regularized finite- K
22 objective whose data term measures the all-fail probability (section 3.1). Fenchel conjugacy in-
23 troduces a per-problem dual state that makes the posterior-dependent term a one-draw expectation
24 (section 3.2). We update this state from running failure estimates through a Bregman update, and its
25 tracking gap measures the difference from exact finite- K optimization (section 3.3).

26 3.1 Posterior learning for fixed sampling budgets

27 Many reasoning problems admit multiple solution paths, so a single solver execution can fail even
28 when another succeeds. Multi-sample inference uses a fixed budget of K executions to increase
29 the chance that the candidate pool contains a successful solution. We make the candidate-generating

30 distribution trainable by learning a posterior $q_\phi(\theta)$ over solver parameters, regularized toward a prior
 31 $p_0(\theta)$. At inference, we draw K parameter samples from this posterior and run each sampled solver
 32 at the prescribed reasoning depth D . The selector associated with the underlying reasoning method
 33 then chooses or aggregates their outputs. We retain that selector and its established training protocol
 34 to isolate posterior learning from selector design. Learning a new selector or co-training the selector
 35 and posterior is a complementary research direction. Our objective targets candidate coverage. The
 36 retained selector determines which candidate the system returns.

37 For problem x , let $F(\theta, x) \in \{0, 1\}$ denote the exact failure indicator for solver θ at depth D , with
 38 $F = 1$ for failure. For independent draws $\theta_{1:K} \stackrel{\text{iid}}{\sim} q_\phi$,

$$\begin{aligned} \Pr(\text{all } K \text{ solvers fail} \mid x) &= \mathbb{E}_{\theta_{1:K}} \left[\prod_{k=1}^K F(\theta_k, x) \right] \\ &= (\mathbb{E}_{\theta \sim q_\phi} [F(\theta, x)])^K. \end{aligned} \quad (1)$$

39 The product is the all-fail indicator. Independence yields the K -th power, so minimizing it maxi-
 40 mizes Pass@ K .

41 Since exact failure is discrete, we use a differentiable failure score $f(\theta, x) \in [0, 1]$. We denote its
 42 posterior mean by

$$\bar{f}_\phi(x) = \mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)]. \quad (2)$$

43 Standard single-solver posterior learning minimizes $\mathbb{E}_x[\bar{f}_\phi(x)]$. Our finite- K posterior objective is

$$\min_{\phi} \mathbb{E}_x[\bar{f}_\phi(x)^K] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0). \quad (3)$$

44 The KL term regularizes the solver posterior toward p_0 [Bissiri et al., 2016, Lacoste-Julien et al.,
 45 2011]. The smooth objective reflects candidate coverage when f agrees with the exact indicator F .

46 3.2 One-draw min-max posterior learning

47 The product of K independent failure scores is an unbiased estimator of $\bar{f}_\phi(x)^K$, but computing it
 48 requires K solver executions per update. A one-draw alternative raises one sampled failure score to
 49 the K -th power. This quantity estimates $\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)^K]$, whereas the finite- K objective contains
 50 $(\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)])^K$. We therefore seek an exact reformulation that avoids K solver executions at
 51 every update.

52 The finite- K objective has an exact min-max form. This form makes the posterior-dependent update
 53 a one-draw expectation for a fixed dual state.

54 Define $\psi_K(\lambda) = (K-1)(\lambda/K)^{K/(K-1)}$ on $[0, K]$. For $K \geq 2$ and $r \in [0, 1]$,

$$r^K = \max_{\lambda \in [0, K]} \{\lambda r - \psi_K(\lambda)\}, \quad \lambda^*(r) = K r^{K-1}. \quad (4)$$

55 Applying eq. (4) to $r = \bar{f}_\phi(x)$ gives the equivalent saddle objective

$$\min_{\phi} \max_{\{\lambda_x \in [0, K]\}_x} \left\{ \mathbb{E}_x [\mathbb{E}_{\theta \sim q_\phi} [\lambda_x f(\theta, x)] - \psi_K(\lambda_x)] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0) \right\}. \quad (5)$$

56 For a finite training set, eq. (5) requires one scalar dual state λ_x for each training problem.

57 For a fixed current value of λ_x , its posterior-dependent term is an ordinary expectation over $\theta \sim q_\phi$.
 58 Assume that q_ϕ admits a differentiable reparameterization and that differentiation can pass through
 59 the expectation. One posterior draw then gives an unbiased stochastic gradient of the fixed-dual term.
 60 The optimal dual state is $\lambda_x^* = K \bar{f}_\phi(x)^{K-1}$. If $\lambda_x = \lambda_x^*$, the one-draw posterior gradient equals the
 61 finite- K data gradient. Otherwise, it remains unbiased for the current fixed-dual objective, whereas it
 62 need not equal the gradient of the original nonlinear finite- K objective. The KL regularizer remains
 63 part of the posterior update.

64 3.3 Bregman update for the dual state

65 The optimal dual state depends on $\bar{f}_\phi(x)$ and changes as the posterior changes. Re-estimating it
 66 from multiple posterior draws at every update would remove the one-draw advantage. Instead, we
 67 update the dual state from the current sampled failure score $f_t = f(\theta_t, x)$.

Let $D_{\psi_K}(\lambda, \mu) = \psi_K(\lambda) - \psi_K(\mu) - \psi'_K(\mu)(\lambda - \mu)$ denote the Bregman divergence generated by ψ_K . We apply the Bregman-proximal ascent step

$$\lambda_{x,t+1} = \arg \max_{\lambda \in [0, K]} \{ \lambda f_t - \psi_K(\lambda) - \eta^{-1} D_{\psi_K}(\lambda, \lambda_{x,t}) \}, \quad \eta > 0. \quad (6)$$

Since $\psi'_K(\lambda) = (\lambda/K)^{1/(K-1)}$, this update has a closed form. With $\beta = 1/(1 + \eta)$ and $\hat{f}_{x,t-1} = \psi'_K(\lambda_{x,t})$,

$$\begin{aligned} \hat{f}_{x,t} &= \beta \hat{f}_{x,t-1} + (1 - \beta) f_t, \\ \lambda_{x,t+1} &= K \hat{f}_{x,t}^{K-1}. \end{aligned} \quad (7)$$

The closed form reduces dual ascent to a scalar exponential moving average (EMA). It requires neither an inner optimization loop nor an additional solver execution. A larger β suppresses sampling noise but increases lag when the posterior failure rate changes.

At update t , we compute $\lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1}$ from the running estimate available before drawing θ_t . We hold this state fixed during the posterior update. Afterward, we incorporate f_t into $\hat{f}_{x,t}$ and use the result at the next update. This order fixes the dual state before the current posterior sample, which gives the conditional one-draw gradient property in section 3.2.

Algorithm 1 One-draw finite- K posterior learning

Require: Posterior q_ϕ , prior p_0 , pool size K , coefficient β , and failure estimates $\{\hat{f}_x\}$
1: **for** each training update t **do**
2: Set and detach $\lambda_{x,t} \leftarrow K \hat{f}_{x,t-1}^{K-1}$ for each minibatch problem x
3: Draw $\theta_t \sim q_\phi$ once and compute $f(\theta_t, x)$
4: Update ϕ with $\lambda_{x,t} f(\theta_t, x)$ and the KL regularizer while holding $\lambda_{x,t}$ fixed
5: Update $\hat{f}_{x,t} \leftarrow \beta \hat{f}_{x,t-1} + (1 - \beta) f(\theta_t, x)$
6: **end for**

Direct finite- K training uses K solver executions per update. The exact min-max reformulation makes the posterior update a one-draw expectation once the current dual state is held fixed. The Bregman step gives a closed-form EMA update for that state without additional solver executions. The practical update follows the finite- K data gradient to the extent that $\hat{f}_{x,t-1}$ tracks $\bar{f}_\phi(x)$.

4 Experiments

4.1 Experimental setup

Datasets. We use Sudoku-Extreme, a set of difficult 9×9 Sudoku completion problems, and Maze-Hard, to evaluate optimal path finding in 30×30 grid mazes. For ARC-AGI-2, the model infers an abstract transformation from a small set of input-output grid pairs and applies it to held-out inputs [Chollet et al., 2025].

Baselines. We compare FB with deterministic reasoning models (HRM, TRM, and FPRM), sampling-based reasoning methods (GRAM, PTRM, W-PTRM, and W-FPRM), and posterior optimization methods (IVON and FB). Detailed method descriptions, initialization, and fine-tuning protocols are provided in appendix B.

4.2 Main results: Finite- K posterior learning improves reasoning

Table 1 shows that FB obtains 99.53% selected accuracy and 99.55% Pass@10 on Sudoku-Extreme. These values exceed W-PTRM, the strongest listed comparator for both metrics, by 0.70 and 0.71 percentage points. On Maze-Hard, FB obtains 88.00% selected accuracy and 91.00% Pass@10. The corresponding gains are 0.30 points over PTRM in selected accuracy and 1.20 points over W-PTRM in Pass@10.

The difference between Pass@10 and selected accuracy is 0.02 points on Sudoku-Extreme and 3.00 points on Maze-Hard. Thus, on Maze-Hard, the candidate pool contains an exact solution for more

Table 1: **FB achieves the highest selected accuracy and Pass@10 on Sudoku-Extreme and Maze-Hard.** Selected accuracy uses each method’s native selector, while Pass@10 measures exact solution coverage among ten candidates. Bold and underlined values mark the best and second-best completed results. [†] All HRM values are paper-reported.

Method	Sudoku-Extreme		Maze-Hard		ARC-AGI-2		
	Acc.	Pass@10	Acc.	Pass@10	Pass@1	Pass@2	Pass@100
HRM [†]	55.00	–	74.50	–	–	5.00	–
FPRM	94.20	–	86.90	–	–	–	–
GRAM	98.48	98.49	87.30	87.30	–	–	–
TRM	87.28	–	87.00	–	2.92	4.58	10.83
PTRM	97.36	97.38	<u>87.70</u>	88.40	2.92	3.75	10.83
W-PTRM	<u>98.83</u>	<u>98.84</u>	87.50	<u>89.80</u>	2.92	3.75	11.25
FB (Ours)	99.53	99.55	88.00	91.00	pending	pending	pending

puzzles than the native selector solves. [RESULT PENDING: We defer the ARC-AGI-2 comparison until the FB evaluation is complete.]

Performance under fixed inference budgets. [PLACEHOLDER: Add the performance- K curves for FB, PTRM, and W-PTRM on Maze-Hard. Report the minimum K at which FB matches each baseline’s $K = 10$ performance and the corresponding reduction in test-time solver executions.]

4.3 Finite- K objective approximation

[PLACEHOLDER: Compare direct K -draw optimization, the naive one-sample surrogate (IVON), and FB. Report the true finite- K objective, the estimated saddle objective, and the dual tracking error over training.]

4.4 Computational efficiency of one-draw training

[PLACEHOLDER: Compare direct K -draw optimization with FB using cumulative solver executions and wall-clock time. Report the finite- K objective or validation Pass@ K against training cost, together with calls per update, time per update, peak memory, and cost to reach a matched target.]

5 Limitations

6 Conclusion

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156 A Finite- K and dual-state derivations

157 A.1 Binary target and direct Monte Carlo estimation

158 Let $p_\phi(x) = \mathbb{E}_{\theta \sim q_\phi}[F(\theta, x)]$ denote the failure probability of one posterior-sampled solver. For
 159 independent draws $\theta_{1:K} \sim q_\phi$,

$$\Pr(\text{all } K \text{ draws fail} \mid x) = \mathbb{E} \left[\prod_{k=1}^K F(\theta_k, x) \right] = p_\phi(x)^K. \quad (8)$$

160 Hence, the probability of at least one success is $1 - p_\phi(x)^K$, which is the corresponding Pass@ K
 161 probability. For binary indicators, the product equals the indicator that all K draws fail. The product
 162 therefore represents the deployment event rather than an additional modeling assumption.

163 Replacing F with the differentiable score f preserves the product across independent draws. Inde-
 164 pendence gives

$$\mathbb{E} \left[\prod_{k=1}^K f(\theta_k, x) \right] = \prod_{k=1}^K \mathbb{E}[f(\theta_k, x)] = \bar{f}_\phi(x)^K. \quad (9)$$

165 Thus, the K -draw product is an unbiased estimator of the finite- K data term. Reusing one draw
 166 estimates $\mathbb{E}[f(\theta, x)^K]$. For $K > 1$, Jensen's inequality gives $\mathbb{E}[f^K] \geq \mathbb{E}[f]^K$, with equality only in
 167 special cases. In the binary limit, $F^K = F$, so the reused-draw objective reduces to the one-draw
 168 failure probability.

169 A.2 Derivation of the Fenchel representation

170 For $K \geq 2$, consider $h_r(\lambda) = \lambda r - \psi_K(\lambda)$ on $[0, K]$. Its derivative on $(0, K)$ is

$$h'_r(\lambda) = r - \left(\frac{\lambda}{K} \right)^{1/(K-1)}. \quad (10)$$

171 The unique stationary point is $\lambda^*(r) = Kr^{K-1}$, which lies in $[0, K]$ for $r \in [0, 1]$. Substitution
 172 gives

$$h_r(\lambda^*(r)) = Kr^{K-1}r - (K-1)r^K = r^K. \quad (11)$$

173 Because ψ_K is strictly convex, h_r is strictly concave and this stationary point is the unique maxi-
 174 mizer for $r \in (0, 1)$. The boundary cases give $\lambda^*(0) = 0$ and $\lambda^*(1) = K$. The case $K = 1$ needs
 175 no Fenchel step because the objective is linear in r .

176 A.3 Fixed-dual one-draw gradient

177 Assume that q_ϕ admits a differentiable reparameterization $\theta = T_\phi(\epsilon)$ whose base distribution does
 178 not depend on ϕ . Also assume that an integrable dominating function permits differentiation under
 179 the expectation. For a fixed detached dual state λ_x ,

$$\begin{aligned} \nabla_\phi \mathbb{E}_{q_\phi}[\lambda_x f(\theta, x)] &= \nabla_\phi \mathbb{E}_\epsilon[\lambda_x f(T_\phi(\epsilon), x)] \\ &= \mathbb{E}_\epsilon[\nabla_\phi \{\lambda_x f(T_\phi(\epsilon), x)\}]. \end{aligned} \quad (12)$$

180 One draw of ϵ , or equivalently one $\theta \sim q_\phi$, therefore gives an unbiased stochastic gradient of the
 181 fixed-dual data term. Moreover,

$$\nabla_\phi \bar{f}_\phi(x)^K = K \bar{f}_\phi(x)^{K-1} \nabla_\phi \bar{f}_\phi(x) = \lambda^*(\bar{f}_\phi(x)) \nabla_\phi \bar{f}_\phi(x). \quad (13)$$

182 The fixed-dual gradient coincides with the finite- K data gradient when $\lambda_x = \lambda^*(\bar{f}_\phi(x))$. For a
 183 lagged dual state, unbiasedness holds conditionally for the fixed-dual objective.

184 A.4 Dual coordinate and Bregman update

185 **Lemma 1** (Dual coordinate). *For integer $K \geq 2$ and $\lambda \in [0, K]$, the derivative $\psi'_K(\lambda) =$
 186 $(\lambda/K)^{1/(K-1)}$ is a strictly increasing bijection from $[0, K]$ to $[0, 1]$. Its inverse is $(\psi'_K)^{-1}(u) =$
 187 Ku^{K-1} for $u \in [0, 1]$, and $\psi'_K(\lambda^*(r)) = r$.*

188 *Proof.* Direct differentiation gives the stated derivative, including its continuous extension at zero.
 189 The derivative increases from zero to one on $[0, K]$. Solving $u = (\lambda/K)^{1/(K-1)}$ gives the inverse,
 190 and substituting $\lambda^*(r) = Kr^{K-1}$ gives the final identity. $\square$

191 **Proposition 1** (Bregman-proximal dual update). *Let $K \geq 2$, $\eta > 0$, $\lambda_{x,t} \in [0, K]$, and $f_t \in [0, 1]$.
 192 If $\hat{f}_{x,t-1} = \psi'_K(\lambda_{x,t})$ and $\beta = 1/(1 + \eta)$, then the unique maximizer in eq. (6) is given by eq. (7).*

193 *Proof.* Expanding the Bregman divergence in eq. (6) and dropping terms independent of λ gives

$$Q_t(\lambda) = \left(f_t + \frac{\hat{f}_{x,t-1}}{\eta} \right) \lambda - \left(1 + \frac{1}{\eta} \right) \psi_K(\lambda) + \text{const.} \quad (14)$$

194 Strict convexity of ψ_K makes Q_t strictly concave. The first-order condition gives

$$\psi'_K(\lambda_{x,t+1}) = \frac{\hat{f}_{x,t-1} + \eta f_t}{1 + \eta} = \beta \hat{f}_{x,t-1} + (1 - \beta) f_t. \quad (15)$$

195 The right-hand side lies in $[0, 1]$, so lemma 1 covers the interior and both boundary cases. Applying
 196 $(\psi'_K)^{-1}$ yields $\lambda_{x,t+1} = K \hat{f}_{x,t}^{K-1}$. $\square$

197 A.5 Indexing and the lagged dual state

198 The implementation follows the order

$$\hat{f}_{x,t-1} \longrightarrow \lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1} \longrightarrow \theta_t \sim q_{\phi_t} \longrightarrow f_{x,t} \longrightarrow \hat{f}_{x,t}. \quad (16)$$

199 Let $\mathcal{H}_{t-1}$ denote the training history before drawing θ_t . The state $\lambda_{x,t}$ is measurable with respect
 200 to $\mathcal{H}_{t-1}$ and is fixed before the current posterior sample. The one-draw gradient result therefore
 201 applies conditionally on this history. Detaching a dual state formed from the current sample blocks
 202 its autograd path but does not make it statistically independent of that sample. Using the lagged state
 203 avoids this dependence.

204 A.6 Stationary stochastic tracking

205 **Proposition 2** (Stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $f_t \in [0, 1]$ satisfy
 206 $\mathbb{E}[f_t \mid \mathcal{H}_{t-1}] = r$ and $\mathbb{E}[(f_t - r)^2 \mid \mathcal{H}_{t-1}] \leq \sigma^2$ for fixed $r \in [0, 1]$. For $\hat{f}_t = \beta \hat{f}_{t-1} + (1 - \beta) f_t$,
 207 $\lambda_{t+1} = K \hat{f}_t^{K-1}$, and $\lambda^* = Kr^{K-1}$,*

$$\mathbb{E}[\hat{f}_t] - r = \beta^t (\hat{f}_0 - r), \quad (17)$$

$$\mathbb{E}[(\hat{f}_t - r)^2] \leq \beta^{2t} (\hat{f}_0 - r)^2 + \sigma^2 \frac{1 - \beta}{1 + \beta} (1 - \beta^{2t}), \quad (18)$$

$$\mathbb{E}[(\lambda_{t+1} - \lambda^*)^2] \leq K^2 (K - 1)^2 \mathbb{E}[(\hat{f}_t - r)^2]. \quad (19)$$

208 *The Fenchel inner-max gap $\Delta_t = r^K - [\lambda_{t+1} r - \psi_K(\lambda_{t+1})]$ satisfies*

$$0 \leq \mathbb{E}[\Delta_t] \leq \frac{K(K-1)}{2} \mathbb{E}[(\hat{f}_t - r)^2]. \quad (20)$$

209 *Proof.* Set $e_t = \hat{f}_t - r$ and $\xi_t = f_t - r$. Then $e_t = \beta e_{t-1} + (1 - \beta) \xi_t$ and $\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0$.
 210 Taking expectations gives eq. (17). Squaring, conditioning, and unrolling the resulting recursion
 211 gives eq. (18). For $g(u) = Ku^{K-1}$ on $[0, 1]$, the bound $|g'(u)| \leq K(K-1)$ and the mean-value
 212 theorem give eq. (19). Finally, $\psi'_K(\lambda^*) = r$ implies $\Delta_t = D_{\psi_K}(\lambda_{t+1}, \lambda^*)$. Writing $u = \hat{f}_t$ and
 213 integrating the derivative of $r^K - Kr u^{K-1} + (K-1)u^K$ between r and u gives eq. (20). $\square$

214 A.7 Non-stationary stochastic tracking

215 **Proposition 3** (Non-stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $r_t = \mathbb{E}[f_t \mid$
 216 $\mathcal{H}_{t-1}]$, $\xi_t = f_t - r_t$, and $\mathbb{E}[\xi_t^2 \mid \mathcal{H}_{t-1}] \leq \sigma_t^2$. Define $e_t = \hat{f}_{t-1} - r_t$, $\lambda_t = K \hat{f}_{t-1}^{K-1}$, and
 217 $\lambda_t^* = K r_t^{K-1}$. Then*

$$e_t = \beta^{t-1} e_1 + (1 - \beta) \sum_{j=1}^{t-1} \beta^{t-1-j} \xi_j - \sum_{j=1}^{t-1} \beta^{t-1-j} (r_{j+1} - r_j). \quad (21)$$

218 *Consequently,*

$$\begin{aligned} \|e_t\|_2 &\leq \beta^{t-1} \|e_1\|_2 + \sum_{j=1}^{t-1} \beta^{t-1-j} \|r_{j+1} - r_j\|_2 \\ &\quad + (1 - \beta) \left(\sum_{j=1}^{t-1} \beta^{2(t-1-j)} \sigma_j^2 \right)^{1/2}. \end{aligned} \quad (22)$$

219 *If $\|r_{t+1} - r_t\|_2 \leq \delta$ and $\sigma_t \leq \sigma$, then*

$$\limsup_{t \rightarrow \infty} \|e_t\|_2 \leq \frac{\delta}{1 - \beta} + \sigma \sqrt{\frac{1 - \beta}{1 + \beta}}. \quad (23)$$

220 *With $\rho_t = \max\{\hat{f}_{t-1}, r_t\}$, the dual error and instantaneous Fenchel gap $\Gamma_t = r_t^K - [\lambda_t r_t - \psi_K(\lambda_t)]$
 221 satisfy*

$$|\lambda_t - \lambda_t^*| \leq K(K - 1) \rho_t^{K-2} |e_t|, \quad (24)$$

$$0 \leq \Gamma_t \leq \frac{K(K - 1)}{2} \rho_t^{K-2} e_t^2. \quad (25)$$

222 *Proof.* The EMA recursion gives $e_{t+1} = \beta e_t + (1 - \beta) \xi_t - (r_{t+1} - r_t)$. Unrolling it yields eq. (21).
 223 The martingale-difference property removes cross terms in the squared norm of the noise sum. The
 224 triangle inequality in L_2 then gives eq. (22). Bounding the drift and noise terms by geometric
 225 series gives eq. (23). Factoring the difference between $\hat{f}_{t-1}^{K-1}$ and r_t^{K-1} gives eq. (24). The identity
 226 $\Gamma_t = D_{\psi_K}(\lambda_t, \lambda_t^*)$ and the same integral argument as in proposition 2 give eq. (25). $\square$

227 The bounds separate sampling noise from drift in the posterior failure rate. Increasing β reduces the
 228 noise term but increases the lag term $\delta/(1 - \beta)$. These results guarantee tracking of the dual state,
 229 not convergence of the full neural-network optimization.

230 B Experimental details

231 B.1 Datasets and evaluation protocols

232 **[PLACEHOLDER:** Report dataset splits, preprocessing, reasoning depth, selectors, the distinc-
 233 tion between K_{train} and K_{eval} , evaluation seeds, and confidence-interval construction for Sudoku-
 234 Extreme, Maze-Hard, and ARC-AGI-2.]

235 B.2 Baseline details

236 **Deterministic reasoning models.** **HRM** couples recurrent modules that operate at different time
 237 scales [Wang et al., 2025]. **TRM** replaces the hierarchy with one small recursive network [Jolicoeur-
 238 Martineau, 2025]. **FPRM** uses fixed-point convergence to halt a looped Transformer [Movahedi
 239 et al., 2026].

Table 2: **FB obtains the highest selected accuracy and Pass@10 with both backbones on Maze-Hard.** Results are percentages on the complete 1,000-puzzle test set. Bold and underlined values mark the best and second-best results within each backbone.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
<i>TRM backbone</i>		
Deterministic	87.00	87.00
PTRM	<u>87.70</u>	88.40
W-PTRM	87.50	<u>89.80</u>
FB (Ours)	88.00	91.00
<i>FPRM backbone</i>		
Deterministic	<u>86.90</u>	—
W-FPRM	<u>86.90</u>	88.70
IVON	84.40	<u>94.00</u>
FB (Ours)	87.70	94.80

Sampling-based reasoning methods. GRAM models reasoning as stochastic latent trajectories and samples several trajectories in parallel [Baek et al., 2026]. PTRM keeps the model parameters fixed, injects Gaussian noise into the latent state at each deep-recursion step, and selects among the resulting trajectories [Sghaier et al., 2026]. W-PTRM and W-FPRM are non-learned baselines that sample each candidate’s parameters from a fixed Gaussian centered at the released TRM or FPRM checkpoint. They add no latent noise.

Posterior optimization methods. IVON (Improved Variational Online Newton) maintains a sampleable diagonal Gaussian through an online variational update and optimizes the single-draw posterior objective [Shen et al., 2024]. FB (Ours) uses the same IVON posterior parameterization and initialization, but optimizes the finite- K objective with Fenchel–Bregman updates. For learned posterior methods, we initialize the posterior mean from a released task checkpoint and fine-tune its mean and diagonal precision on the task training split. Fine-tuning adapts the solver distribution to the chosen objective, while W-PTRM and W-FPRM isolate sampling without this adaptation.

B.3 Training and compute accounting

[PLACEHOLDER: Report optimization hyperparameters, posterior and dual-state initialization, training-update budgets, hardware, batch sizes, mixed-precision settings, solver-call accounting, wall-clock measurement, and peak-memory measurement.]

C Additional posterior analyses

C.1 Posterior learning across recursive backbones

Table 2 compares posterior methods with TRM and FPRM backbones on the complete Maze-Hard test set. Every stochastic method uses ten candidates ($K = 10$) and its native selector.

FB improves both metrics under the two tested backbones. With TRM, it raises selected accuracy by 0.30 percentage points over PTRM and Pass@10 by 1.20 points over W-PTRM. With FPRM, FB improves selected accuracy from 84.40% to 87.70% and Pass@10 from 94.00% to 94.80% over IVON. The paired 95% bootstrap intervals are [1.60, 5.10] for selected accuracy and [0.00, 1.60] for Pass@10. The two FPRM posterior rows use the same initialization, training data, update budget, and evaluation noise.

C.2 Posterior construction under a fixed backbone

Table 3 fixes the FPRM backbone and Maze-Hard protocol while varying how each method constructs its parameter distribution. Diagonal Laplace fits a local Gaussian from curvature at a point estimate [Daxberger et al., 2021], while SWAG estimates a Gaussian from an SGD trajectory [Mad-dox et al., 2019].

Table 3: The comparison fixes FPRM and Maze-Hard while varying the posterior construction.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
FPRM	86.90	–
W-FPRM	86.90	88.70
Diagonal Laplace	pending	pending
SWAG	pending	pending
IVON	84.40	94.00
FB (Ours)	87.70	94.80

Among posterior results, FB improves selected accuracy by 3.30 percentage points and Pass@10 by 0.80 points over IVON. [RESULT PENDING: Compare FB with Diagonal Laplace and SWAG after their matched Maze-Hard results and fit times are available.]

D Additional objective and resource analyses

D.1 Extended objective and tracking diagnostics

[PLACEHOLDER: Report the naive one-sample surrogate, detailed saddle-objective gap, per-problem and aggregate dual tracking errors, and sensitivity of the tracking gap over training.]

D.2 Extended inference-budget results

[PLACEHOLDER: Report selected accuracy over K , additional sampling budgets, per-seed results, and confidence intervals complementing the main performance- K curves.]

E Ablation studies

[PLACEHOLDER: Report ablations for the EMA coefficient β , KL coefficient τ_{KL} , dual-state initialization, and training budget K_{train} . Include final finite- K objective, tracking error, Pass@ K , and selected accuracy where applicable.]